

Zhendong ANG

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<https://ang9876.github.io/>

EDUCATION

- 2022.08 - present **National University of Singapore, Singapore**
Ph.D. in Computer Science
Adviser: Prof. Umang Mathur
- 2019.09 - 2022.07 **École Polytechnique Fédérale de Lausanne, Switzerland**
M.Sc. in Computer Science
- 2015.09 - 2019.06 **Zhejiang University, P. R. China**
B.Sc. in Computer Science

INTERNSHIP

- 2025.08 - 2025.11 **Amazon Web Services**
2024.08 - 2024.11 **Amazon Web Services**
2021.07 - 2022.01 **Oracle Labs, Zürich**

PUBLICATIONS

- Under Submission **Zhendong Ang**, Azadeh Farzan, Umang Mathur, “Enhanced Data Race Prediction Through Modular Reasoning”
- LMPL 2025 Mike He, **Zhendong Ang**, Ankush Desai, Aarti Gupta, “Ranking Formal Specifications using LLM”
- CAV 2024 **Zhendong Ang**, Umang Mathur. “Predictive Monitoring with Strong Trace Prefixes.”
- POPL 2024 **Zhendong Ang**, Umang Mathur. “Predictive Monitoring against Pattern Regular Languages.”
- CPP 2024 Qiyuan Zhao, George Pîrlea, **Zhendong Ang**, Umang Mathur, Ilya Sergey. “Rooting for Efficiency: Mechanised Reasoning about Array-Based Trees in Separation Logic.”
2024 CPP Distinguished Paper Award

AWARDS

- 2025/2026 NUS School of Computing Research Achievement Award
2023/2024 NUS School of Computing Research Achievement Award

PROJECTS

- 2022.02 - 2022.07 **Macro Annotations for Scala 3 (EPFL LAMP – Programming Methods Laboratory)**
- Designed patterns and APIs for macro annotations in Scala 3.
 - Implemented macro annotations expansion.
- 2021.03 - 2021.07 **ScalaPy Related Projects (EPFL LAMP – Programming Methods Laboratory)**
- Created Numpy interface in Scala using ScalaPy and explored pattern of type facades.
 - Migrated ScalaPy to Scala 3: refactored the code and used macros in Scala 3.
 - Supported exposing custom Scala types to Python.
- 2020.09 - 2021.01 **Extending GADTs support in Scala 3 (EPFL LAMP – Programming Methods Laboratory)**
- Explored how subtype, class type parameters, and GADTs work in Dotty.
 - Added GADTs constraint to class type parameters.
 - This new feature has been merged into Scala 3.

TALKS

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| 2024.07 | Predictive Monitoring with Strong Trace Prefixes. | CAV 2024 |
| 2024.01 | Predictive Monitoring against Pattern Regular Languages. | POPL 2024 |
| 2023.07 | New Results of Membership Problems for Trace Languages. | Highlights of Logic, Games and Automata 2023 |